SIMATS ENGINEERING
Department of Artificial Intelligence and Data Science

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE AI	**Date:**	18/08/26

Code of Conduct

*I **A. Mounish / 192472273** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the Student: ___________________________

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and

interpreting the data.

- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- Active(x) → Producing(x)
- Produces(x,y)∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines. An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language

"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Solutions:

1. Action–object relationships

Query	Action	Object / Service	Entity
Activate international roaming for my number.	ACTIVATE	Roaming	Customer
Deactivate caller tune service.	DEACTIVATE	CallerTune	Customer
Check my data balance.	QUERY	DataBalance	Customer
Enable 5G service.	ACTIVATE	5GService	Customer

Interpretation: ACTIVATE(Roaming, Customer) means that the customer wants the roaming service activated. Similarly, DEACTIVATE(CallerTune, Customer) represents removal of the caller-tune service, while QUERY(DataBalance, Customer) requests information rather than a service change. ACTIVATE(5GService, Customer) represents enabling 5G service.

Action	Object / Service	Entity
ACTIVATE	Roaming	Customer
DEACTIVATE	Caller Tune	Customer
QUERY	Data Balance	Customer

Figure 1. Action–object–entity relationships derived from the supplied semantic representations.

2. Semantic interpretation error

Q2 contains the semantic interpretation error in the chatbot log. The actual intent is "Deactivate Caller Tune", whereas the predicted intent is "Activate Caller Tune". The representation DEACTIVATE(CallerTune, Customer) is therefore semantically correct, but the chatbot's final intent classification is incorrect.

Intent accuracy = 3 correct predictions / 4 total predictions × 100 = 75%

Intent error rate = 1 incorrect prediction / 4 total predictions × 100 = 25%

3. Effect of meaning representation on chatbot decision accuracy

- A correct semantic representation preserves the user's intended action and the target service.
- The distinction between ACTIVATE and DEACTIVATE is critical because the two actions produce opposite service outcomes.
- If negation or action polarity is lost during semantic interpretation, the chatbot may execute an unwanted service operation.
- In the supplied data, one incorrect intent prediction reduces decision accuracy to 75%, demonstrating the practical effect of semantic errors.

4. Recommended improvements

- Use intent classification with explicit action labels such as ACTIVATE, DEACTIVATE, QUERY and similar operational categories.
- Add negation and polarity detection so that words such as "deactivate", "disable" and "cancel" are not confused with activation requests.
- Use slot/entity extraction to separately identify the service, customer and other required parameters.
- Use confidence thresholds and human-agent fallback for uncertain predictions.
- Continuously evaluate the classifier using confusion matrices and real customer-chat logs, with special attention to opposite-action errors.

Conclusion

The semantic representation is useful because it converts natural-language requests into structured actions and entities. The supplied chatbot data shows that Q2 is the only incorrect prediction, giving 75% intent accuracy. Improving action polarity, entity extraction, context handling and confidence-based validation would make telecom chatbot decisions more reliable.

QUESTION 2 – First-Order Predicate Calculus for Smart Manufacturing

The manufacturing scenario provides four machine states and three logical rules. First-Order Predicate Calculus (FOPC) can represent these facts and derive machine-production states systematically.

1. Production data represented using FOPC

$$\mathbf{Active(M1)}$$

$$\mathbf{Active(M2)}$$

$$\mathbf{Maintenance(M3)}$$

$$\mathbf{Active(M4)}$$

Given predicate rules

$$\mathbf{Active(x) \rightarrow Producing(x)}$$

$$\mathbf{Produces(x,y) \land Active(x) \rightarrow Available(y)}$$

$$\mathbf{Maintenance(x) \rightarrow \neg Producing(x)}$$

2. Logical inference about machine production

- From Active(M1) and Active(x) → Producing(x), we infer Producing(M1).
- From Active(M2) and Active(x) → Producing(x), we infer Producing(M2).
- From Active(M4) and Active(x) → Producing(x), we infer Producing(M4).
- From Maintenance(M3) and Maintenance(x) → ¬Producing(x), we infer ¬Producing(M3).

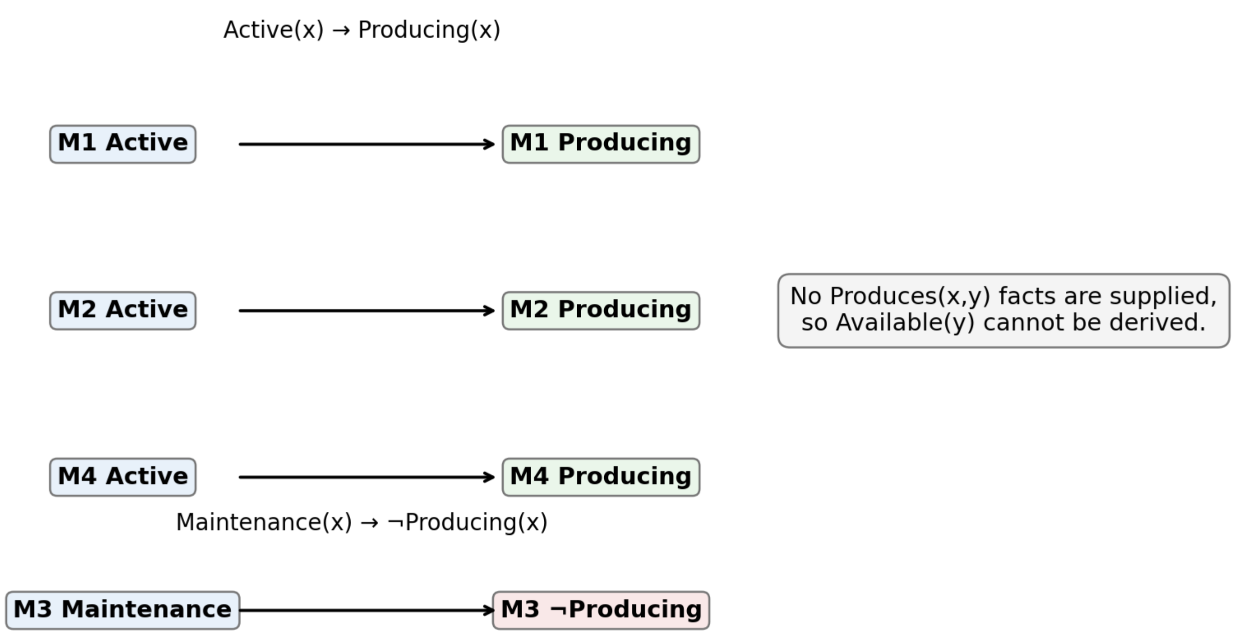

Figure 2. Predicate-based inference from the machine-status data.

3. Products currently available

The rule for product availability is Produces(x,y) ∧ Active(x) → Available(y). However, the supplied production data does not provide any Produces(x,y) facts linking a machine to a particular product. Therefore, no specific product can be logically declared Available(y) from the given information alone.

Hence, the correct data-supported conclusion is: product availability cannot be determined from the supplied facts. Additional information such as Produces(M1,Gear) would be required before Available(Gear) could be inferred.

4. Effect of maintenance on Gear production

The supplied data establishes that M3 is under maintenance and therefore M3 is not producing. However, the question does not state that M3 produces Gear. Consequently, FOPC does not permit a definite conclusion that Gear production is affected by M3's maintenance. If an additional fact such as Produces(M3,Gear) were provided, then the maintenance state would logically indicate that M3 cannot continue producing Gear.

5. Effectiveness of predicate logic in industrial decision-support

- FOPC represents machine states and relationships in a precise, structured form.
- Rules make the reasoning process explicit and explainable, which is useful for industrial monitoring.
- The system can automatically derive consequences such as producing or not producing from machine states.
- Its limitation in this scenario is incomplete information: without machine-to-product relationships, product availability cannot be inferred.
- Therefore, predicate logic is effective when the underlying facts and rules are complete, accurate and consistently maintained.

Conclusion

FOPC provides a clear mechanism for converting manufacturing status data into logical conclusions. From the supplied data, M1, M2 and M4 are producing, while M3 is not producing. No specific available product or Gear-production impact can be concluded without the missing Produces(x,y) relationship.

QUESTION 3 – Word Sense Disambiguation in E-Commerce Search Engines

The supplied search logs contain ambiguous words with multiple possible meanings. The clicked-result information provides direct contextual evidence for selecting the intended sense.

1. Correct word sense for each query

User Query	Possible Ambiguity	Context / Clicked Result	Correct Sense
Apple accessories	Fruit / Technology Brand	iPhone Charger	Technology Brand
Mouse wireless	Animal / Computer Device	Bluetooth Mouse	Computer Device
Java tutorial	Island / Programming Language	Coding Lessons	Programming Language
Python course	Snake / Programming Language	Software Development Training	Programming Language

Query	Selected Sense	Context / Clicked Result
Apple accessories	Technology / Brand	iPhone Charger
Mouse wireless	Computer Device	Bluetooth Mouse
Java tutorial	Programming Language	Coding Lessons

Figure 3. Context-based word sense disambiguation using the supplied clicked results.

2. Semantic cues used for disambiguation

- Query context: words such as “accessories”, “wireless”, “tutorial” and “course” narrow the intended meaning.
- Clicked-result evidence: iPhone Charger, Bluetooth Mouse, Coding Lessons and Software Development Training directly indicate the selected sense.
- Domain context: the e-commerce/search environment increases the likelihood of product or technology meanings.
- Co-occurrence information: related terms appearing with an ambiguous word provide useful semantic clues.
- User interaction signals can provide feedback for ranking the most relevant sense.

3. Impact of incorrect sense selection

- An incorrect sense can return irrelevant search results and reduce user satisfaction.
- Recommendation systems may suggest unrelated products or content.
- Search relevance and click-through performance may decrease because the results do not match the user’s actual intention.
- Repeated sense errors can reduce trust in the search engine.

4. Industrial-scale WSD strategy

- Generate candidate senses from a lexical knowledge source and domain-specific product taxonomy.
- Encode the query together with surrounding context using a context-aware language model or semantic embedding model.
- Combine textual similarity with clicked-result history and other interaction signals.
- Rank candidate senses using a confidence score and send low-confidence cases to a fallback or broader search strategy.
- Continuously retrain and evaluate the WSD component using search logs, click feedback and newly emerging product terminology.
- Maintain domain-specific vocabularies so that technology terms are correctly interpreted in e-commerce contexts.

Conclusion

The four supplied queries can be disambiguated successfully using contextual and clicked-result evidence. Correct WSD improves search relevance and recommendation quality, while incorrect sense selection can directly reduce the usefulness of an e-commerce search system.

QUESTION 4 – Syntax-Driven Semantic Analysis in Healthcare Information Systems

The healthcare scenario shows how syntactic structure can guide semantic role assignment. The supplied system uses Subject–Verb–Object parsing, while the semantic role table assigns roles such as Agent, Instrument, Recipient and Symptom.

1. Contribution of syntactic structure to semantic role assignment

- The subject generally identifies the entity performing or experiencing the action.
- The verb identifies the main event or action in the sentence.
- The object identifies the entity affected by the action, while prepositional phrases can provide additional roles such as recipient.
- Correct parsing therefore provides the structural information required for semantic role labeling.

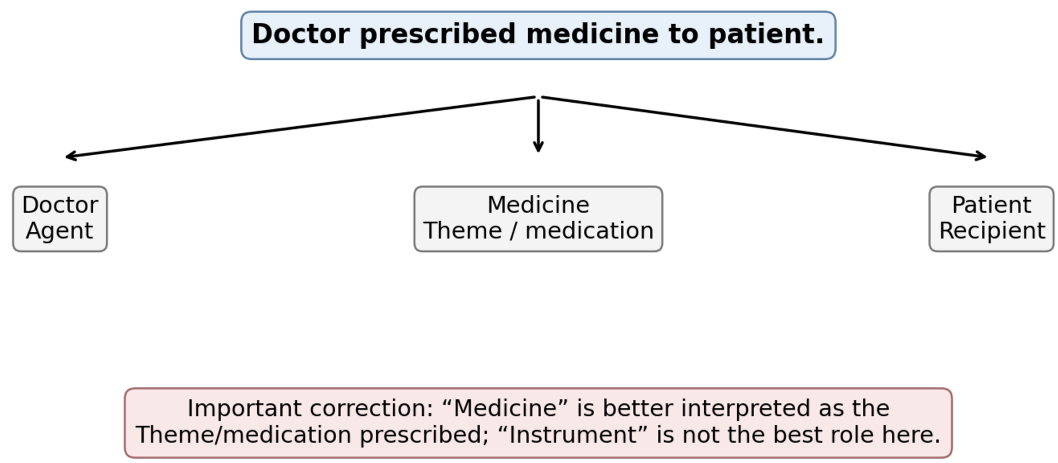

Figure 4. Semantic-role interpretation of the first healthcare sentence.

2. Evaluation of the assigned semantic roles

Sentence / Entity	Role in supplied table	Evaluation	Reason
Doctor	Agent	Appropriate	Doctor performs the prescribing action.
Medicine	Instrument	Needs correction	In "prescribed medicine", medicine is better treated as the Theme/medication being prescribed, not the instrument used to

			perform the action.
Patient	Recipient	Appropriate	Patient receives the prescribed medicine.
Headache	Symptom	Appropriate	A severe headache represents a reported medical symptom.

For the remaining sentences, “Patient” in “Patient reported severe headache” is the experiencer/reporter, “Nurse” in “Nurse monitored patient continuously” is the Agent, and “Medicine” in “Medicine reduced blood pressure” functions as the cause/agent-like entity of the reduction event. These relationships show why syntax must be interpreted together with the meaning of the verb and the medical context.

3. Potential errors caused by incorrect parsing

- Subject–object reversal can assign the action to the wrong person or entity.
- Incorrect attachment of phrases such as “to patient” can change the recipient or relationship between entities.
- A wrong semantic role can confuse medication, patient, symptom and clinician relationships.
- Negation or modifier errors can change the clinical meaning of a statement.
- In healthcare applications, such errors can affect information extraction, clinical search, summarization and decision-support outputs.

4. Recommended methods to improve syntax-driven semantic analysis

- Use dependency parsing in addition to simple Subject–Verb–Object patterns so that grammatical relationships are represented more accurately.
- Apply semantic role labeling to identify Agent, Theme, Recipient, Experiencer and other clinically relevant roles.
- Use healthcare-domain language models and medical terminology resources to improve interpretation of clinical vocabulary.
- Add rule-based validation for critical relationships such as medication–patient, symptom–patient and clinician–action links.
- Evaluate the parser and semantic-role system on annotated clinical sentences and review high-risk errors before deployment.

Conclusion

Syntax provides the structural foundation for semantic interpretation, but syntax alone is not always sufficient. The supplied example correctly identifies several roles, while the assignment of Medicine as an Instrument is less appropriate for the verb “prescribed”. Combining syntactic parsing, semantic role labeling and healthcare-specific validation can produce more reliable clinical NLP results.

Overall Assessment Insight

Across the four tasks, the central requirement is to interpret data rather than reproduce definitions. The strongest answers therefore connect each supplied fact to a semantic representation, logical inference, contextual cue or syntactic relationship, and clearly distinguish conclusions that are supported by the given data from conclusions that require additional information.